Rearranging Formulas

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Simple rearranging

What are formulas?

- A **formula** is a rule, definition or relationship between different quantities, written in shorthand using **letters** (**variables**)
  - Formulas include an **equals** sign
- Some examples you should be **familiar** with are:
  - The equation of a straight line
    - $y = mx + c$
  - The area of a trapezium
    - $\text{Area} = \frac{(a + b)h}{2}$
  - Pythagoras' theorem
    - $a^2 + b^2 = c^2$

How do I rearrange formulas?

- The letter (**variable**) that is on its **own** on one side is called the **subject**
  - y is the subject of $y = mx + c$
- To make a different letter the subject, we need to **rearrange the formula**
  - This is also called **changing the subject**
- The method is as follows:
  - First, **remove** any **fractions**
    - Multiply both sides by the lowest common denominator
  - Then use **inverse** (opposite) **operations** to get the variable on its own
    - This is similar to solving equations
- For example, make x the subject of $\frac{5x + 6}{2} = y$
  - First remove fractions
    - Multiply both sides by 2
      $5x + 6 = 2y$
  - Then get x on its own
    - Subtract 6 from both sides
      $5x = 2y - 6$

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- There may be more than one correct way to write an answer
  - The following are acceptable **alternative** forms
    $x = \frac{2y}{5} - \frac{6}{5}$
    $x = \frac{2(y-3)}{5}$
    $x = 0.4(y-3)$
    $x = 0.4y - 1.2$

Should I expand brackets?

- If the variable is inside the brackets, you can either:
  - **expand** the brackets
  - or **divide** both by the coefficient if it is not zero
- For example, make x the subject of $3(1+x) = y$
  - You can expand and then rearrange
    - $3 + 3x = y$
    - $3x = y - 3$
    - $x = \frac{y-3}{3}$
  - You can divide and then rearrange
    - $1 + x = \frac{y}{3}$
    - $x = \frac{y}{3} - 1$
  - Both answers are equivalent
    - $\frac{y-3}{3}$ is the same as $\frac{y}{3} - 1$

Examiner Tips and Tricks

Stop and think when you see a bracket. If the variable you are trying to make the subject is not inside the bracket, then you do not need to expand.

For example, to make x the subject of $(1+k)x = y$, you can simply divide both sides by $(1+k)$ to get $x = \frac{y}{1+k}$.

What if I get fractions in fractions?

- Some rearrangements can lead to **fractions in fractions**
  - $x = \frac{\frac{3}{t}}{2}$
- Either rewrite with a **divide sign**, $\div$, then use the method of **dividing two fractions**

$$x = \frac{3}{t} \times \frac{1}{2}$$

$$x = \frac{3}{2t}$$

- Or **multiply top and bottom** by the the **lowest common denominator** of the two fractions and **cancel**
  - $x = \dfrac{\frac{5}{y}}{\frac{t}{8}}$ becomes $x = \dfrac{\frac{5}{y} \times 8y}{\frac{t}{8} \times 8y} = \dfrac{40}{ty}$

What if I end up dividing by a negative?

- Remember that $\dfrac{a}{-b}$ (minus **below**) is the same as $\dfrac{-a}{b}$ (minus **above**) and the same as $-\dfrac{a}{b}$ (minus **outside**)
  - Though be careful, as $\dfrac{-a}{-b}$ is $\dfrac{a}{b}$
- $-2x = y - 3$ becomes $x = \dfrac{y-3}{-2}$ (minus **below**)
  - This is the same as $x = \dfrac{-(y-3)}{2}$ (minus **above**) or $-\dfrac{y-3}{2}$ (minus **outside**)
    - **brackets** are **required** for minus **above**
    - **brackets** are **assumed** for minus **outside**
  - You can also **expand** the brackets $\dfrac{-(y-3)}{2} = \dfrac{-y+3}{2} = \dfrac{3-y}{2}$

Examiner Tips and Tricks

- Mark schemes will accept different forms of the same answer, as long as they are correct and fully simplified.

Worked Example

Make x the subject of the following.

(a) $4m + 5x = 3$

Answer:

Get $5x$ on its own by subtracting $4m$ from both sides

$$5x = 3 - 4m$$

Get x on its own by dividing both sides by 5

(b) $3t = \frac{2}{x}$

Answer:

Remove fractions by multiplying both sides by the denominator, x

$$3tx = 2$$

Get x on its own by dividing both sides by $3t$

$$x = \frac{2}{3t}$$

(c) $A = \frac{9(1-4x)}{2g}$

Answer:

Remove fractions by multiplying both sides by the denominator, $2g$

$$2gA = 9(1-4x)$$

x is inside the brackets
Expand the brackets to release the x term

$$2gA = 9 - 36x$$

One way to get x on its own is by subtracting 9 then dividing by -36
Or you can first add $36x$ to both sides, to create positive $36x$ on the left

$$2gA + 36x = 9$$

Now get x on its own by subtracting $2gA$ then dividing by 36

$$36x = 9 - 2gA$$
$$x = \frac{9-2gA}{36}$$

$$x = \frac{9-2gA}{36}$$

Other accepted forms of the answer are:

$$\frac{2gA-9}{-36}, \quad \frac{-(2gA-9)}{36}, \quad -\frac{2gA-9}{36}, \quad \frac{1}{4} - \frac{gA}{18}$$

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